Multi-subject Identification of Hand Movements Using Machine Learning

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Abstract. Electromyographic (EMG) signals provide information about muscle activity. In hand movements, each gesture's execution involves the activation of different combinations of the forearm muscles, which generate distinct electrical patterns. Furthermore, the analysis of muscle activation patterns represented by EMG signals allows recognizing these gestures. We aimed to develop an automatic hand gesture recognition system based on supervised Machine Learning (ML) techniques. We trained eight computational models to recognize six hand gestures and generalize between different subjects using raw data recordings of EMG signals from 36 subjects. We found that the random forest model and fully connected artificial neural network showed the best performances, indicated by 96.25% and 96.09% accuracy, respectively. These results improve on computational time and resources by avoiding data preprocessing operations and model generalization capabilities by including data from a larger number of subjects. In addition to the application in the health sector, in the context of Smart Cities, digital inclusion should be aimed at individuals with physical disabilities, with which, this model could contribute to the development of identification and interaction devices that can emulate the movement of hands.

Keywords: Activity recognition · Computational modeling · Electromyography · Machine learning

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1 Introduction

Electromyography (EMG) signals are generated by the electrical activity of skeletal muscle fibers and can be captured through two ways. First, invasive methods involve the insertion of a needle directly into the muscle; second, non-invasive methods rely on surface electrodes that are placed on the skin in areas close to the target muscle. In particular, surface EMG is the most suited for establishing human-machine interfaces of routine use [1]. These kind of signals are random, highly non-stationary [2], and display subtle variations associated with anatomical and motor coordination differences between subjects [3,4]. Therefore, different techniques of feature extraction and selection for time series [5] must be applied, which are also useful for developing optimal models of artificial intelligence (AI) [6].

These signals are used by health professionals for several purposes, including disease diagnosis and rehabilitation. Recently, with the increase number of automatic learning techniques, these signals are used in systems for movement classification [7,8], recognition of motor unit action potential (MUAP) [9], and diagnosis of neuromuscular diseases [10]. These tasks are achieved by measuring and studying features that can be extracted from EMG signals. Feature extraction can be done through different techniques, such as autoregressive models [11], signal entropy measurements [12], and statistical measurements of amplitude in the time and frequency domains [6]. This last technique includes the widely used Wavelet transform, which provides information in the frequency domain at high and low frequencies [13].

The feature extraction stage is followed by the classification stage, which relies on models that can learn from the extracted features [14]. Among computational models, AI is considered the main domain, which includes Machine Learning (ML) and Deep Learning (DL). AI refers to the process of displaying human intelligence features on a machine. ML is used to achieve this task, while DL is a set of models and algorithms used to implement ML [15]. In general, ML and DL technologies are powerful for extracting features and finding relationships between data; therefore, these approaches are suitable for tasks that rely on taking advantage of human experience [16,17]. Machine and Deep Learning algorithms applied to electrophysiological signals constitute a rapidly growing field of research and allow researchers to train computer systems as experts that can be used later on to support decision-making processes. From ML, the most commonly used models in this field are Support Vector Machine (SVM) [18], K-Nearest Neighbors (KNN) [9], and Ensemble classifiers [19,20]; in DL, different architectures are reported, including fully connected artificial neural networks (ANN) and convolutional neural networks (CNN) [21].

This paper proposes an approach to EMG signal classification, in which not all information of the time series is considered. Here, only raw data for hand gestures (i.e., the muscular activation values provided by each channel) are used to identify the patterns associated with six gestures, regardless of the subject performing the action. Therefore, we aimed to generalize among different subjects, which is a current issue in this classification task. The data used in this research consists of EMG recordings from 36 patients performing six different hand gestures. The classification accuracy obtained using this approach is over 90% for several models, demonstrating feasibility and outperforming state-of-the-art classification results involving several patients.

This paper is organized as follows: Sect. 2 presents the materials and methods, including the database and models used in the proposed framework, which are explained in detail in Sect. 3. Section 4 provides the experimental results and discussion. Finally, Sect. 5 concludes the paper and raises issues for further research.

2 Materials and Methods

2.1 Database

The information used to develop this research was retrieved from a free database called *EMG data for gestures Data Set.* available on the website of the UCI Machine Learning Repository. This database contains surface EMG signal recordings in a time interval where six different static hand gestures are executed, providing a total of 72 recordings from 36 patients. The static hand gestures included resting hand, grasped hand, wrist flexion, wrist extension, ulnar deviation, and radial deviation. These gestures were performed for three seconds, each with a three-second interval between gestures. The information was collected using the *MYO thalmic bracelet* device, which has eight channels equally spaced around the patient's forearm. Accordingly, each recording consists of an eight-channel time series, in which each segment is properly labeled with numbers zero to six, where zero corresponds to the intervals between gestures and numbers one to six refer to the gestures mentioned above, as reported in Table 1. This database was consolidated by [3] during their research on the factors limiting human-machine interfaces via sEMG.

Table 1. Labels of the gestures contained in the database.

Label	Gesture
0	Intervals between gestures
1	Resting hand
2	Grasped hand
3	Wrist flexion
4	Wrist extension
5	Ulnar deviation
6	Radial deviation

2.2 Machine Learning Models

The ML models implemented in the experiments included K-Nearest Neighbors (KNN), Logistic Regression (LR), Gaussian Naive Bayes (GNB), Multilayer Perceptron (MLP) using one hidden layer, Random Forest (RF), and Decision Tree (DT).

2.3 Deep Learning Models

Fully Connected Artificial Neural Network Architecture. The model described here corresponds to a network with three hidden layers and one output layer. The output layer has six units corresponding to the number of movements to identify. The activation function for the output layer was softmax and the loss function was categorical cross-entropy. The diagram of the ANN architecture is shown in Fig. 1.

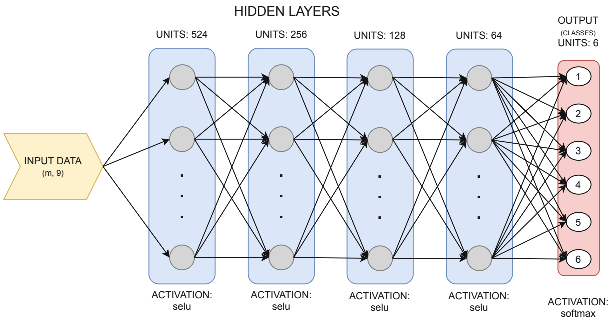


Fig. 1. Architecture of the fully connected artificial neural network used in this study.

Convolutional Neural Network Architecture. The structure of the model proposed here is a convolutional neural network with two 1D convolutional layers, one ReLU activation layer, three fully connected layers, and one output layer. The output layer has six units corresponding to the number of movements to identify. The activation function for the output layer was softmax and the loss function was categorical cross-entropy. The diagram for the CNN architecture is presented in Fig. 2.

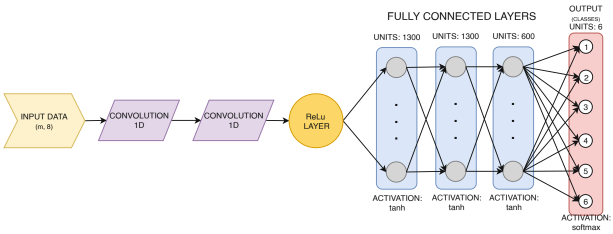


Fig. 2. Convolutional neural network architecture used in this study.

2.4 Tools

The complete experimental procedure is implemented using Google Colab platform and Python v. 3.7.3 programming language. Database import and preprocessing were done with Pandas library v. 0.24.2. The ML models mentioned in Sect. 2.2, in addition to GridSearchCV, principal component analysis (PCA), train-test split, and the scaling function were implemented using Scikit-learn library v. 0.21.2 [22]. The artificial neural network architectures and the corresponding training and validation processes were implemented using Keras library v. 2.2.4 [23]. Further, the resources used in this research are available in a public GitHub repository containing the EMG recordings files from the database, as well as a Jupyter notebook with the implemented source code. The repository is available here.

3 Proposed Framework

Figure 3 shows the steps of the proposed framework. Each step is explained in detail in the following subsections.

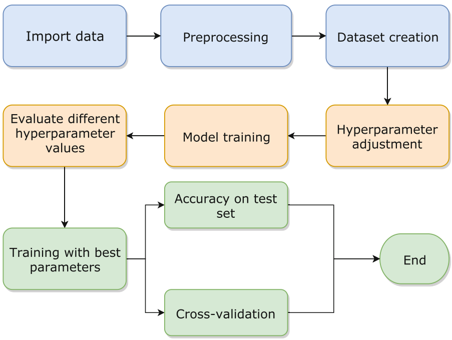


Fig. 3. Flowchart of the proposed framework.

3.1 Importing the Data

The data from the database is presented as a plain text file (.txt). The *read_csv* function from Pandas library was used to import these files.

3.2 Preprocessing

Data labeled as zero was removed since these corresponded to the resting intervals between gesture execution and do not contain relevant information to solve the problem. Therefore, only data captured during the execution of the six gestures of interest were kept. No feature extraction procedure was implemented in order to use the values of the eight channels provided by the device as discriminating information between gestures (i.e., raw data).

Considering the high susceptibility of EMG signals to noise and external artifacts, as well as the fact that relevant information of the signal is located in the frequency spectrum range from 0 450 Hz [24], we applied three different digital filters to eliminate or reduce noise as much as possible. The first filter consisted of a high pass filter with cut-off frequency 20 Hz, which eliminated most of the low-frequency noise while preserving the majority of the signal information [24]. The second and third filters were band pass with cut-off frequencies of 20–180 Hz and 10–480 Hz, respectively; this type of filter is commonly used to remove low-frequency noise and high-frequency information that may be irrelevant to the classification task [7,8].

3.3 Dataset Creation

For dataset creation, the patients were correctly distributed into training, validation, and test sets. We verified that each patient was present in only one set to avoid bias induced by including data from the same patient during training and testing. The training set comprised concatenated data from 30 patients, the validation set included three patients, and test set contained the remaining three patients. The patients in each of the sets were randomly selected. A second group of sets was created by scaling the first one; therefore, all features equally contributed to the training process. The sets described above were used during the hyperparameter adjustment process and to compute the *test set accuracy* metric. Finally, to conserve the correct patient distribution during the cross-validation process, each of the nine k-folds was manually generated by concatenating the data from four randomly selected patients.

3.4 Hyperparameter Adjustment

After dataset creation, we adjusted the parameters for each classification method. Given the number of variable parameters for ANN and CNN architectures, hyperparameter adjustment was done by tuning one or two parameters at a time using the GridSearchCV function implemented in Scikit-learn library. This method selects the best combination of parameters using the result of a cross-validation process. The parameters were adjusted in the following order: batch_size and epochs, optimization algorithm, network weight initialization, activation function, and neurons in hidden layers.

3.5 Model Training and Hyperparameter Optimization

For parameter optimization, we used an approach similar to the grid search. Given a set of parameters and value ranges, the model was trained and tested on the validation set using every possible combination. For traditional ML models, this process was implemented manually using *for* loops, whereas, for artificial neural networks, it was done using the GridSearchCV function from Scikit-learn library due to the high number of parameters.

3.6 Model Evaluation Using the Best Parameters

After performing hyperparameter optimization, we considered the best set of parameters for each model to generate the definitive evaluation metric. For this, cross-validation was used to measure model performance. This method uses the complete feature matrix and divides the dataset into k-fold parts. In this case, k-folds = 9 and each part contained data from four randomly selected patients. The data was iterated over these partitions, using eight for training and one for validation, until all of these passed through both states. This procedure was used to check the stability of the results. In addition to cross-validation, the test accuracy provided an estimate of generalization, given that the test set had not been used in any previous stage of the experiment. Finally, we computed precision, recall, and F1-score as per class performance measures.

To test for statistically significant differences in the cross-validation results applied to the raw and processed data among the models, we performed an analysis of one-way variance test (ANOVA) (i.e., under the assumption of normality) with the following structure:

$$H_0 : \mu_1 = \mu_2 = \mu_3 = \mu_4$$

$$H_A : \mu_i \neq \mu_j \ for \ i \neq j \wedge i = 1, ..., 4$$

Where μ_i is the mean of each dataset (raw data, 20 Hz, bandpass 20–180 Hz, bandpass 10–480 Hz).

4 Experiments and Discussions

4.1 Results

The best parameter values and datasets (normal/scaling) for each ML model were selected based on the results of the parameter optimization step (Table 2). Similarly, for ANN and CNN architectures, the optimized parameter values with GridSearchCV are shown in Table 3, and the neural networks were trained using these values. Performance accuracy and loss curves were generated in each training epoch to observe model behavior and determine the performance of the learning process. For CNN, the number of epochs was manually adjusted according to the model training curves; as a result, the most appropriate value found was Epochs = 100. Figure 4 shows the accuracy and loss curves during the training process for ANN.

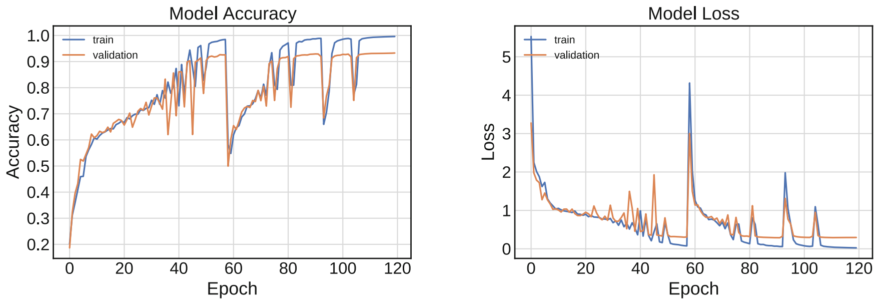
The two methods mentioned in Sect. 3 for measuring model performance, namely cross-validation and test set accuracy, were implemented using the optimized parameters for each model. The cross-validation results for each model and filtered data are shown in Table 4, where STD stands for standard deviation.

Table 2. ML models

Models	Best parameter value
KNN	n_neighbors = 1
	Dataset: Normal
LR	C = 0,3
	Dataset: Scaling
GNB	N/A
MLP	hidden_layer_sizes = 800
	Dataset: Scaling
RF	n_estimators = 71
	Dataset: Scaling
DT	Default value:
	max_depth = None
	min_samples_split = 2
	max_features = None
	Dataset: Scaling

Table 3. ANN and CNN

Parameters	Optimal values	
	ANN	CNN
Batch size	50,000	10,000
Epochs	120	100
Optimization algorithm	lr = 0.01	lr = 0.001
Weight initialization	Normal	Normal
Activation function	tanh	Linear tanh
Units in hidden layers	1,050	1,300, 1,300, 600
Filters	N/A	128
Kernel_size	N/A	3



(a) (b)

Fig. 4. Training curves using raw data - ANN

In summary, the results from Table 4 indicate that LR, GNB, and MLP models display the lowest performance. On the other hand, the best results were achieved by KNN, RF, DT, ANN and CNN (93.22%, 95.39%, 93.29%, 93.73%, 94.77%, respectively). The ANOVA test for the cross-validation data did not show statistically significant differences between raw and processed data (P-value = 0.994 at a significance level of 5%). Test set accuracy was computed for the best-performing models using raw data and per class performance measures (e.g., precision, recall and F1-score). The resting hand gesture showed the best classification according to the established measures. The results per class for the top three models are reported in Table 5.

Table 4. Performance of the models based on cross-validation using different filters.

Models	Cross-validation (Accuracy [%] ± STD [%])			
	Raw data	Highpass 20 Hz	Bandpass 20–180 Hz	Bandpass 10-480 Hz
K-Nearest Neighbors	95.06 ± 1.98	94.94 ± 2.72	95.31 ± 2.69	95.03 ± 1.81
Logistic Regression	22.25 ± 3.34	16.81 ± 0.20	17.39 ± 0.20	17.62 ± 0.52
Gaussian Naive Bayes	63.27 ± 1.09	50.12 ± 1.31	58.46 ± 1.67	60.68 ± 1.22
Multilayer Perceptron	72.68 ± 1.11	62.95 ± 1.34	67.24 ± 1.32	68.40 ± 0.99
Random Forest	96.25 ± 1.63	95.78 ± 2.29	96.13 ± 2.32	96.02 ± 1.50
Decision Tree	94.59 ± 2.08	94.86 ± 2.86	95.13 ± 2.78	94.80 ± 1.89
Artificial Neural Network	96.09 ± 1.74	93.55 ± 3.85	95.59 ± 2.76	95.51 ± 1.64
Convolutional Neural Network	95.84 ± 1.81	94.64 ± 2.49	94.12 ± 1.06	93.82 ± 0.99

Table 5. Model performance based on class types.

Class	Model								
	RF			ANN			CNN		
	Precision	Recall	F1-score	Precision	Recall	F1-score	Precision	Recall	F1-score
Resting hand	0.93	1.00	0.96	0.93	0.99	0.96	0.94	1.00	0.97
Grasped hand	0.98	0.92	0.95	0.96	0.91	0.94	0.97	0.92	0.95
Wrist flexion	0.93	0.97	0.95	0.93	0.95	0.94	0.94	0.96	0.95
Wrist extension	0.97	0.96	0.96	0.94	0.94	0.94	0.95	0.95	0.95
Ulnar deviation	0.95	0.91	0.93	0.93	0.88	0.90	0.95	0.89	0.92
Radial deviation	0.96	0.96	0.96	0.94	0.94	0.94	0.94	0.95	0.95

4.2 Discussions

In this study, we explored the performance of different ML and DL techniques in the task of classifying six hand gestures from electromyographic signals. We found no statistically significant differences generated by using different preprocessing digital filters; thus, these filters did not benefit model performance compared to the raw data. We recommend working with raw data, without affecting performance, to save time and computational resources that would be invested in the preprocessing stage. Our reports demonstrate the feasibility of using the information provided by each of the channels as discriminant features between gestures. The best results described here are comparable to literature reports on similar problems and the same classification task. For example, [25] reported an accuracy of 99.2% in the diagnosis of neuromuscular disorders using data from 27 subjects, and a rotation forest classifier. Furthermore, [18] used EMG signals from lower limbs of two subjects to classify two different gait phases (e.g., swing and stance), reporting a classification accuracy of 96% using a SVM classifier. Both of these previous studies report a high classification accuracy; however, each considered the classification task of only two classes.

On the other hand, [21] reported an accuracy of 98.12% for the identification of seven hand gestures for a single subject using four time-domain features and stacked sparse autoencoders as a classifier. Similarly, [26] achieved a 96.38% accuracy using a RF model to identify three hand gestures in a subject-independent experiment with data from 10 subjects. Overall, these previous studies report a high classification accuracy; however, they also considered a small number of subjects. Furthermore, it is important to determine how the transition from single to multi-subject classification affects precision. Table 6 provides a summary of the works described.

Table 6. Performance comparison with other related works.

Methods	Task	Classes	Subjects	Accuracy (%)
Subasi et al. [25]	Neuromuscular disorders	2	27	99.2
Zieger et al. [18]	Gait phases	2	2	96
Zia ur Rehmann et al. [21]	Gestures	7	1	98.12
Wahid et al. [26]	Gestures	3	10	96.38
Rabin et al. [27]	Gestures	6	1–5	94.8–77.3
Our method	Gestures	6	36	96.25

In this context, we demonstrated the feasibility of traditional ML and DL models to identify different hand gestures from raw EMG signals. Furthermore, we achieved generalization among 36 subjects, while maintaining a high classification accuracy. Given our approach, each record can be used as an instance of the features matrix to obtain a more significant amount of labeled information for model training. Also, this approach avoids computational costs and reduces the time required to implement the different feature extraction methods. For instance, using Google Colab platform in a GPU environment, the KNN model takes less than a minute for training and testing; RF takes approximately seven minutes, and both DL models take nearly 20 min. These computational time metrics are not provided by other authors, which makes it difficult to determine if our approach is faster.

The accuracy obtained for Random Forest (96.25%), ANN (96.09%) and CNN (95.84%) (Table 4) is comparable to the literature reports previously discussed and outperforms the results presented as state-of-the-art for this classification task. Furthermore, our results correspond to a cross-validation procedure that provides robustness to the model and demonstrates the consistency of the results. The dataset used in this research (i.e., containing data from 36 patients) is bigger than those previously used in most multi-subject EMG signal classification tasks. Unlike other studies, we considered the correct distribution of the patients into the training, validation, and test sets to limit the information of a patient to only one dataset. This is an important point when designing and implementing ML models since these can memorize the information in the training set and, if the same information is in the validation or test set, biased performance measures may be generated.

5 Conclusions

The results confirm the feasibility of using machine learning and deep learning techniques to identify muscle activation patterns, specifically, hand movements.

Further, this paper presents a different approach to this classification problem using raw data provided by the *MYO thalmic bracelet* device to train models with high performance levels, regardless of the subject to which the signals belong. This allows overcoming different anatomical and biological factors between subjects, which translate into subtle differences in EMG signals that previously limited the performance of classification systems for different subjects. The models achieved high accuracy in multi-subject classification with 36 subjects. The best performances were obtained by implementing Random Forest with 96.25% accuracy, artificial neural network with 96.09%, convolutional neural network with 95.84%, K-nearest neighbor with 95.06%, and decision tree with 94.59%. Future work should be aimed to optimize and adapt the trained models for real-time classification tasks, such as motor control for prostheses.

References

1. Phinyomark, A., Khushaba, R.N., Scheme, E.: Feature extraction and selection for myoelectric control based on wearable EMG sensors. Sensors **18**(5), 1615 (2018)
2. Phinyomark, A., Campbell, E., Scheme, E.: Surface electromyography (EMG) signal processing, classification, and practical considerations. In: Naik, G. (eds.) Biomedical Signal Processing. SERBIOENG, pp. 3–29. Springer, Singapore (2020). https://doi.org/10.1007/978-981-13-9097-5_1
3. Lobov, S., Krilova, N., Kastalskiy, I., Kazantsev, V., Makarov, V.: Latent factors limiting the performance of sEMG-interfaces. Sensors **18**(4), 1122 (2018)
4. Molla, M.K.I., Shiam, A.A., Islam, M.R., Tanaka, T.: Discriminative feature selection-based motor imagery classification using EEG signal. IEEE Access **8**, 98255–98265 (2020)
5. Shakeel, A., Tanaka, T., Kitajo, K.: Time-series prediction of the oscillatory phase of EEG signals using the least mean square algorithm-based AR model. Appl. Sci. **10**(10), 3616 (2020)
6. Phinyomark, A., Phukpattaranont, P., Limsakul, C.: Feature reduction and selection for EMG signal classification. Expert Syst. Appl. **39**(8), 7420–7431 (2012)
7. Phukan, N., Kakoty, N.M., Shivam, P., Gan, J.Q.: Finger movements recognition using minimally redundant features of wavelet denoised EMG. Health Technol. **9**, 579–593 (2019). https://doi.org/10.1007/s12553-019-00338-z
8. Ji, Y., Sun, S., Xie, H.-B.: Stationary wavelet-based two-directional two-dimensional principal component analysis for EMG signal classification. Meas. Sci. Rev. **17**(3), 117–124 (2017)
9. Jahromi, M.G., Parsaei, H., Zamani, A., Stashuk, D.W.: Cross comparison of motor unit potential features used in EMG signal decomposition. IEEE Trans. Neural Syst. Rehabil. Eng. **26**(5), 1017–1025 (2018)
10. Diaz, M., Ferrer, M.A., Impedovo, D., Pirlo, G., Vessio, G.: Dynamically enhanced static handwriting representation for Parkinson's disease detection. Pattern Recogn. Lett. **128**, 204–210 (2019)
11. Koçer, S., Tümer, A.E.: Classifying neuromuscular diseases using artificial neural networks with applied autoregressive and cepstral analysis. Neural Comput. Appl. **28**(1), 945–952 (2016). https://doi.org/10.1007/s00521-016-2383-8
12. Chen, W., Wang, Z., Xie, H., Yu, W.: Characterization of surface EMG signal based on fuzzy entropy. IEEE Trans. Neural Syst. Rehabil. Eng. **15**(2), 266–272 (2007)

13. Gokgoz, E., Subasi, A.: Comparison of decision tree algorithms for EMG signal classification using DWT. Biomed. Sig. Process. Control **18**, 138–144 (2015)
14. Awad, A.I., Hassaballah, M.: Image Feature Detectors and Descriptors: Foundations and Applications. SCI, vol. 630. Springer, Cham (2016). https://doi.org/10.1007/978-3-319-28854-3
15. Bengio, Y., Goodfellow, I.J., Courville, A.: Deep learning. Technical report (2015)
16. Orozco-Arias, S., Isaza, G., Guyot, R., Tabares-Soto, R.: A systematic review of the application of machine learning in the detection and classification of transposable elements. PeerJ **7**, e8311 (2019)
17. Tabares-Soto, R., Raúl, R.P., Gustavo, I.: Deep learning applied to steganalysis of digital images: a systematic review. IEEE Access **7**, 68970–68990 (2019)
18. Ziegier, J., Gattringer, H., Mueller, A.: Classification of gait phases based on bilateral EMG data using support vector machines. In: Proceedings of the IEEE RAS and EMBS International Conference on Biomedical Robotics and Biomechatronics, Augus, vol. 2018, pp. 978–983. IEEE Computer Society, October 2018
19. Subasi, A., Yaman, E., Somaily, Y., Alynabawi, H.A., Alobaidi, F., Altheibani, S.: Automated EMG signal classification for diagnosis of neuromuscular disorders using DWT and bagging. Procedia Comput. Sci. **140**, 230–237 (2018)
20. Benazzouz, A., Guilal, R., Amirouche, F., Hadj Slimane, Z.E.: EMG feature selection for diagnosis of neuromuscular disorders. In: 2019 International Conference on Networking and Advanced Systems (ICNAS), pp. 1–5 (2019)
21. Rehman, M.Z.U., et al.: Multiday EMG-based classification of hand motions with deep learning techniques. Sensors **18**(8), 2497 (2018)
22. Pedregosa, F., et al.: Scikit-learn: machine learning in Python. J. Mach. Learn. Res. **12**, 2825–2830 (2011)
23. Chollet, F., et al.: Keras (2015). https://keras.io
24. De Luca, C.J., Gilmore, L.D., Kuznetsov, M., Roy, S.H.: Filtering the surface EMG signal: movement artifact and baseline noise contamination. J. Biomech. **43**(8), 1573–1579 (2010)
25. Subasi, A., Yaman, E.: EMG signal classification using discrete wavelet transform and rotation forest. In: Badnjevic, A., Škrbić, R., Gurbeta Pokvić, L. (eds.) CMBEBIH 2019. IP, vol. 73, pp. 29–35. Springer, Cham (2020). https://doi.org/10.1007/978-3-030-17971-7_5
26. Wahid, M.F., Tafreshi, R., Al-Sowaidi, M., Langari, R.: Subject-independent hand gesture recognition using normalization and machine learning algorithms. J. Comput. Sci. **27**, 69–76 (2018)
27. Rabin, N., Kahlon, M., Malayev, S., Ratnovsky, A.: Classification of human hand movements based on EMG signals using nonlinear dimensionality reduction and data fusion techniques. Expert Syst. Appl. **149**, 113281 (2020)